Artificial Professional Coder

Proposed production model · Retrieve, rerank, select

Apply the principles of professional occupation coding to open-ended, self-reported survey answers. The system proposes a four-digit unit code from the CBS national taxonomy, based on ISCO-08. The supplied draft contains 465 selectable codes; the catalogue is versioned.

1 Source-backed occupation cards

1. Four-digit code, official title and hierarchy: major, sub-major, minor and unit groups.
2. Source-backed definitions across relevant hierarchy levels, typical duties and occupational context.
3. Detailed occupations and role variants, with aliases linked to the appropriate occupation.
4. Inclusion/exclusion criteria, coding rules and examples of boundary cases.
5. **Expert-verified survey examples: de-identified answers and confirmed codes from a separate, validated survey sample.**

Every card records its sources and version. Card examples are separate from the held-out gold standard.

2 Analytical workflow · one survey record

Prepared survey answers · the same query is supplied to all three stages

Taxonomy text and aliases ↓	Occupation cards ↓	Occupation cards + rules ↓
RETRIEVE	**RERANK**	**SELECT**
Text embeddings + cosine similarity	Score each query/card pair independently	Compare all M cards jointly under coding rules
Top K · 50 codes	**Top M · 10 codes**	**One proposed code**

→ →

Validate the proposed code → accept under a tested policy,
or route to an occupation coder; insufficient information can remain unresolved.

K=50 and M=10 are starting values, tuned on development data. Reranker scores measure relevance; they are not calibrated probabilities of a correct code.

3 Independent gold-standard validation set

Use representative, recent surveys independently coded by two experts, with disagreements adjudicated. Keep each evaluation version frozen; validated additions create a new version. Keep its cases and related duplicates out of cards and tuning. Report Recall@K/Recall@M, final accuracy/Macro F1, automatic coverage and accepted-code accuracy, review workload, and time/cost per 1,000 cases.

Release improvement: experts curate cards using the separate example pool. Tune K, M and acceptance rules on development/validation data; approve changes after independent evaluation against the baseline. Choose open-weight or enterprise models by Hebrew quality, privacy, context limits and measured cost.

Production proposal · Verified-example enrichment, a real-data gold set and review routing are proposed additions.

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github.com/david-shomron/LLM-BATCH-CLASSIFICATION